

IINTS-AF Physiology Reference Brochure

Competition evidence bundle for EUCYS review

Project: IINTS-AF SDK

Purpose: Give doctors, engineers, and judges one compact document that explains which physiological quantities the SDK models, which numbers matter, and how to read them correctly.

Audience: Clinical reviewers, biomedical engineers, EUCYS judges.

This PDF was rendered from the markdown report and packaged benchmark artifacts inside the IINTS-AF SDK workspace.

Why This Document Exists

Glucose simulation is easy to make visually convincing and hard to make scientifically honest. A curve can look smooth and still be physiologically weak. IINTS-AF therefore separates four ideas that are often mixed together:

Layer	Question it answers
physiology	what happens inside the virtual patient?
measurement	what does the CGM-like sensor report?
interpretation	how do we summarize the trace?
protection	when does the safety layer block or stop a run?

The SDK is strongest when those layers remain explicit rather than hidden inside one opaque score.

The Core Physiological Story

Type 1 diabetes simulation in IINTS-AF revolves around a small set of interacting quantities:

Quantity	Unit	Physiological role
glucose	mg/dL	primary state variable shown in plots and reports
insulin-on-board (IOB)	U	insulin still active after delivery
carbs-on-board (COB)	g	carbohydrate still being absorbed
basal insulin rate	U/h	background insulin exposure
insulin sensitivity (ISF)	mg/dL/U	expected glucose effect of 1 U insulin
insulin-to-carb ratio (ICR)	g/U	carbohydrate amount covered by 1 U insulin
glucose trend	mg/dL/min	direction and steepness of change
exercise intensity	0-1	extra glucose-lowering pressure

These quantities let the SDK represent common research questions:

- what happens after a meal?
- what happens if insulin stacks?

- what changes during exercise?
- what if the announced carbohydrates are wrong?
- what if the sensor is delayed or noisy?

Numbers With Clinical Meaning

These numbers are used for interpretation, not as proof that a simulation is clinically validated.

Quantity	Value	Why it matters
target glucose range	70-180 mg/dL	standard CGM Time-in-Range band
below-range threshold	<70 mg/dL	low-glucose exposure
level-2 hypoglycemia	<54 mg/dL	more serious low-glucose state
common adult TIR target	>70%	interpretable reporting benchmark
common adult time-below-range target	<4% below 70 mg/dL	about 58 minutes/day or less
common adult severe-low target	<1% below 54 mg/dL	about 15 minutes/day or less

The SDK reports these values because they make results interpretable across studies. It does not claim that meeting one table row proves real-world safety.

Software Safety Rails

The safety supervisor uses intentionally conservative software limits. These are not personalized treatment prescriptions.

Safety rail	SDK default	Meaning inside the software
max single bolus	5.0 U	blocks an oversized one-step insulin request
max insulin per hour	3.0 U	limits recent cumulative delivery
max insulin on board	4.0 U	limits active insulin burden
falling-trend stop	-2.0 mg/dL/min	avoids dosing into a rapid fall
critical stop	<40 mg/dL for 30 min	terminates an unsafe synthetic trajectory
plausible sensor range	40-500 mg/dL	guards impossible CGM-like input

This separation is useful for judges: a physiological threshold says what glucose means; a supervisor threshold says what the research software is allowed to do.

What A Meaningful Day Looks Like

A realistic virtual day should not be a decorative sine wave. It should tell a plausible physiological story.

Event	Example time	Example magnitude	Expected effect
breakfast	07:30	48 g carbs	first post-meal rise
lunch	12:15	62 g carbs	larger midday disturbance
exercise	12:45	intensity 0.35 for 30 min	lowers glucose after lunch
dinner	18:00	74 g carbs	strongest evening challenge
snack	21:30	18 g carbs	small late excursion

That table matters because it links the plotted glucose curve to causes a doctor or engineer can reason about.

Starter Profiles

The SDK now ships with explicit first-use profiles instead of leaving beginners with ambiguous defaults.

Profile	Initial glucose	Basal	ISF	ICR	Drift	Best use
stable-demo	130 mg/dL	0.2 U/h	40 mg/dL/U	15 g/U	0.001	demos and smoke tests
stress-test	120 mg/dL	0.5 U/h	50 mg/dL/U	10 g/U	0.003	harder disturbances
endurance	140 mg/dL	0.0 U/h	20 mg/dL/U	25 g/U	0.0	long unattended software runs

The most important design lesson is that a harmless-looking coefficient such as a glucose-decay setting can completely change the trajectory. A simulator is only as honest as its defaults.

Measurement Is Not Physiology

Algorithms usually see a CGM-like signal, not perfect latent glucose. IINTS-AF therefore models sensor behavior separately.

Sensor profile	Noise	Lag	Typical use
ideal	0 mg/dL	0 min	debugging
clinical_cgm	7 mg/dL	10 min	clean realistic baseline
free_living_cgm	8 mg/dL	10 min	free-living realism
compression_prone	8.5 mg/dL	12 min	artifact-heavy stress testing

This makes an important scientific distinction possible:

- true glucose can be physiologically stable
- the reported glucose can still be noisy, delayed, or temporarily wrong

Model Families

Model family	Strength	Best use
simplified custom model	transparent, fast, stress-test friendly	demos, regressions, safety sweeps
Bergman-style model	more mechanistic internal states	physiology-focused experiments
empirical residual layer	adds day-scale irregularity	realism studies and synthetic mirrors

No single model is "the real patient." The scientific goal is to keep assumptions visible and compareable.

What The SDK Still Does Not Fully Model

Limitation	Current status
glucagon and counter-regulatory hormones	not explicit
illness, infection, steroids, menstrual-cycle effects	scenario-level approximation only
mixed fat/protein meal kinetics	only coarse delayed-meal approximations
formulation-specific insulin pharmacokinetics	simplified configurable timing
individualized real-patient digital twins	not claimed

Those omissions are not hidden defects; they define the boundary of what the SDK can honestly claim today.

How To Explain The SDK In One Minute

1. IINTS-AF does not dose real patients; it is a research simulator.
2. It models glucose, insulin action, meals, exercise, and CGM imperfections separately.
3. It reports standard CGM metrics such as Time in Range and hypoglycemia exposure.

4. It wraps algorithms in a deterministic safety supervisor so risky outputs become visible and auditable.
5. It is useful because it turns insulin-algorithm development from "one pretty graph" into reproducible evidence.

Best One-Sentence Claim

IINTS-AF is a pre-clinical research SDK that keeps virtual-patient physiology, CGM measurement behavior, and safety supervision explicit so insulin algorithms can be tested more honestly before any real-world claim is made.

Source Backbone

This brochure follows the maintained source legend in docs/EVIDENCE_BASE.md, including the ADA Standards of Care 2026, the international Time-in-Range consensus, OhioT1DM, Dalla Man/Cobelli meal-model literature, UVA/Padova simulator literature, and the 2024 Communications Medicine paper on generative T1D simulation.